9.1-2 $Q_4 = \sum_{i=1}^{5} \frac{(Y_i - np_i)^2}{np_i} = \frac{(224-580(0.4))^2}{580(0.4)} + \frac{(119-580(0.2))^2}{580(0.2)}$

$+ \frac{(130-580(0.2))^2}{580(0.2)} + \frac{(48-580(0.1))^2}{580(0.1)} + \frac{(59-580(0.1))^2}{580(0.1)}$

$= 3.784$. Since $\chi^2_{0.05}(4) = 9.488 > 3.784$, this test fails at 5% significance to reject H_0.

9.1-6 $p_{00} = P(X=0) = \binom{3}{0}(.3)^0(0.7)^3 = 0.343$; $p_{10} = \binom{3}{1}(0.3)(0.7)^2 = 0.441$; $p_{20} = \binom{3}{2}(0.3)^2(0.7) = 0.189$; $p_{30} = \binom{3}{3}(0.3)^3(0.7)^0 = 0.027$. $\rightsquigarrow Q_3 = \frac{(57-200(0.343))^2}{200(0.343)} +$

$\frac{(95-200(0.441))^2}{200(0.441)} + \frac{(38-200(0.189))^2}{200(0.189)} + \frac{(10-200(0.027))^2}{200(0.027)} = 6.405$.

Since $\chi^2_{0.1}(3) = 6.251 < 6.405 < 7.815 = \chi^2_{0.05}(3)$, the p-value is between 0.05 and 0.1.

Let Y = ~~total~~ # of heads. Then $Y = 1\cdot 95 + 2\cdot 38 + 3\cdot 10 = 201$. Using formula (7.3-2), an (approximate) 95% confidence interval for p is $\hat{p} \pm z_{\alpha/2}\sqrt{\frac{\hat{p}(1-\hat{p})}{n}} = \frac{201}{600} \pm 1.96\sqrt{\frac{\frac{201}{600}\left(\frac{399}{600}\right)}{600}}$

$\approx \boxed{0.335 \pm 0.0378}$.

9.2-3 If $A_1=[0,56]$, $A_2=(56,75.5]$, $A_3=(75.5,100]$, then we get the table

	A_1	A_2	A_3
Class U	7	4	4
Class V	3	6	6

Estimate of p_i $(i=1,2,3)$ is $10/30=1/3$ $\rightarrow$ multiply by 15, get 5. So Q is

$$q=\frac{(7-5)^2}{5}+\frac{(4-5)^2}{5}+\frac{(4-5)^2}{5}+\frac{(3-5)^2}{5}+\frac{(6-5)^2}{5}+\frac{(6-5)^2}{5}$$

$$=\frac{12}{5}=2.4<5.991=\chi^2_{0.05}(3-1)=\chi^2_{0.05}(2).$$ $\rightarrow$ Don't reject H_0.

9.2-4 Dividing combined sample into $A_1=[0,61.5]$, $A_2=(61.5,78.5]$, $A_3=(78.5,100]$, table is

	A_1	A_2	A_3
Class U	9	4	2
Class V	5	5	5
Class W	1	6	8

Estimate $p_i=\frac{15}{45}=1/3$; multiply by 15, get 5. So Q is

$$q=\frac{1}{5}\left[(9-5)^2+(4-5)^2+(2-5)^2+(5-5)^2+(5-5)^2+(5-5)^2+(1-5)^2+(6-5)^2+(8-5)^2\right]=\frac{52}{5}=10.4>9.488=\chi^2_{0.05}(4)=$$

$$\chi^2_{0.05}((\overset{h}{3}-1)(\overset{k}{3}-1)).$$ So reject H_0.

9.2-13 (c) We have $\hat{p}_{1\cdot}=\frac{73}{426}$ (grad school), $\hat{p}_{2\cdot}=\frac{239}{426}$ (high school), $\hat{p}_{3\cdot}=\frac{114}{426}$ (college), & $\hat{p}_{\cdot 1}=\frac{188}{426}$ (newspaper), $\hat{p}_{\cdot 2}=\frac{189}{426}$ (TV), $\hat{p}_{\cdot 3}=\frac{49}{426}$ (radio). So Q

$$q = \sum_{j=1}^{h} \sum_{i=1}^{k} \frac{[y_{ij} - n\hat{p}_{i\cdot}\hat{p}_{\cdot j}]^2}{n\hat{p}_{i\cdot}\hat{p}_{\cdot j}} = \frac{\left(45 - 426\left(\frac{73}{426}\right)\left(\frac{188}{426}\right)\right)^2}{426\left(\frac{73}{426}\right)\left(\frac{188}{426}\right)} + \dots$$

$$+ \frac{\left(13 - 426\left(\frac{114}{426}\right)\left(\frac{49}{426}\right)\right)^2}{426\left(\frac{114}{426}\right)\left(\frac{49}{426}\right)} = 11.398 > 9.488 = \chi^2_{0.05}(4) \Rightarrow \text{reject } H_0$$

(i.e. reject independence of education & media credibility).
From table IV, p-value is between 0.01 and 0.025.
(following course notes online)

$\boxed{7.}$ (a) We assume $X_{ij} \sim N(\mu_i, \sigma^2)$, where X_{ij} = mpg of brand "i" ($1 \leftrightarrow A$, $2 \leftrightarrow B$, $3 \leftrightarrow C$), on j^{th} trial/test. $H_0: \mu_1 = \mu_2 = \mu_3$; H_1: not H_0, i.e. not all three means are equal.

Test statistic is $F := \dfrac{SS(T)/(m-1)}{SS(E)/(n-m)}$, $m = 3$, $n = 12$.

Critical region: $\{F \geq F_\alpha(m-1, n-m)\} = \{F \geq F_{0.05}(2,9) = 4.26\}$.

(b) $SS(TO) = \sum_{i=1}^{m} \sum_{j=1}^{n_i} X_{ij}^2 - \frac{1}{n}\left(\sum_i \sum_j X_{ij}\right)^2 = 16.969$,

$SS(T) = \sum_{i=1}^{m} \frac{1}{n_i}\left(\sum_{j=1}^{n_i} X_{ij}\right)^2 - \frac{1}{n}\left(\sum_i \sum_j X_{ij}\right)^2 =$
11.783, $SS(E) = SS(TO) - SS(T) = 5.186$. $\Rightarrow F = \left(\frac{12-3}{3-1}\right) \times \left(\frac{11.783}{5.186}\right) = 10.225 > 4.26. \Rightarrow$ Reject H_0 (i.e., brands have different performance levels).

(c) As $F_{0.01}(2,9) = 8.02$, p-value is < 0.01. (Exact: 0.0048)

	Deg. freedom	MS	F
SS(T)	2	5.8915	10.225
SS(E)	9	0.5762	
SS(TO)	11		

$\boxed{9.1\text{-}13}$ We use 8 classes $A_1, \dots, A_8$ with equal probability $1/8 = 0.125$. $\bar{x} = 320.1$, $s^2 = 45.56$. $\Rightarrow$ the class boundaries are 320.1, $320.1 \pm \Phi^{-1}(0.375) \cdot s$, $320.1 \pm \Phi^{-1}(0.25) \cdot s$, $320.1 \pm \Phi^{-1}(0.125) \cdot s$, or

$$-\infty < 312.338 < 315.551 < 317.947 < 320.1 < 322.253 < 324.649 < 327.862 < \infty.$$

For each class, the expected frequency is $50/8 = 6.25$. The observed frequencies are

A_1	A_2	A_3	A_4	A_5	A_6	A_7	A_8
7	5	5	10	4	6	5	8

Therefore $q = \big[(7-6.25)^2 + (5-6.25)^2 + (5-6.25)^2 + (10-6.25)^2 + (4-6.25)^2 + (6-6.25)^2 + (5-6.25)^2 + (8-6.25)^2\big]/6.25 = 4.4 < 11.07 = \chi^2_{0.05}(5)$, so <u>don't reject</u>. (We use $\chi^2_{0.05}(5)$ because we estimated two parameters, μ & σ^2, & $8-1-2=5$.)